Distant Close Reading — Technical Appendix

This appendix documents every rule, prompt, run and measure behind the Stage 1 proposal, in the order the pipeline runs. Full prompt text, code and outputs are in the repository.

A1. Corpus selection

Documents were selected by a fixed rule run against OpenAlex (`select_corpus.py`), written and committed before any reading. A candidate must (1) cite at least one anchor work, (2) be an article in one of six journals, (3) contain an include term in its title or abstract, and (4) contain no exclude term. Candidates are ranked by total citations and the top N selected; anchors are added where specified.

Journals (by ISSN): *Academy of Management Review* (0363-7425), *Academy of Management Journal* (0001-4273), *Administrative Science Quarterly* (0001-8392), *Organization Science* (1047-7039), *Strategic Management Journal* (0143-2095), *Journal of Management Studies* (0022-2380).

Corpus	Anchors (OpenAlex ID)	Include terms	Exclude terms	Citing articles	Pass rule	Selected
Legitimacy	Suchman 1995 (W2162412465); Deephouse & Suchman 2008 (W2496218594); Suddaby, Bitektine & Haack 2017 (W4240547422)	legitimacy	none	511	181	20 + 3 anchors
TCE	Williamson 1975 (W1973308529), 1981 (W1972698480), 1985 (W1684540946), 1991 (W2109280325)	transaction cost; make-or-buy; make or buy	none	1,136	172	15 + 4 anchors
Splice	Jensen & Meckling 1976 (W2752617332)	corporate governance; board(s) of directors	transaction cost	804	140	5

No anchor reached the cap of 1,000 citing articles, so every candidate pool is complete. Anchor IDs were resolved once, checked by hand and then locked. The two Williamson books are indexed in OpenAlex as review records carrying the books' citations; they were used for the citation lookup, and the books themselves were read.

Version log.

- *v1 (adopted, frozen 2026-10-21):* the rule above.
- *v2 (considered and rejected the same day, before any reading):* title-only matching, with the Williamson books used for lookup only. It was prompted by the v1 TCE list being dominated by critics and rival theories. It was rejected because contestation is not incoherence: critics who characterize a construct the same way are evidence of coherence, and papers that merely invoke a construct show how it is used. The rejection, and its reason, are recorded in the script.

Out of scope: Granovetter (1985) and other critiques published outside the six journals; adding journals after the rule was fixed would have changed all three corpora.

Document sources. PDFs were obtained through JSTOR, Google Scholar links and ResearchGate (author uploads). Because copies from different sources can differ (cover pages, scans, preprint versus published versions), the extraction manifest stores a hash of each document's extracted text, so a replicator can tell whether their copy matches. One article (W2124927217) was replaced with a JSTOR copy with its cover page removed; see A2.

A2. Passage extraction

Passages were selected by a fixed rule (`extract_passages.py`, PyMuPDF), identical for every document:

1. Extract text block by block; join hyphenated line breaks; drop blocks under 40 words (headers, footers, captions).
2. Drop everything from the first "References" or "Bibliography" heading in the second half of the document.
3. Keep paragraphs that match at least one construct term for the corpus.
4. Rank by number of term matches (ties: earlier first) and keep at most 8 per document, so no document dominates.

Corpus	Construct terms (regular expressions)
Legitimacy	`legitim\w*`
TCE and splice	`transaction[- ]cost`, `asset specificit`, `specific assets?`, `opportunis`, `bounded(ly)? rational`, `governance`, `hierarch`, `make[- ]or[- ]buy`, `vertical(ly)? integrat`, `hybrid`

Splice documents use the TCE terms because they are read as part of the TCE corpus; "governance" is the term they share in a different sense.

Books. For the two Williamson books, the rule takes the original preface or introduction plus Chapter 1, cut between two marker phrases recorded in the script and manifest. Later prefaces (paperback and college editions) are excluded. *Markets and Hierarchies* was available only as a scanned-book text file without paragraph breaks, so its passages are whole pages (about 400 words, against 80–160 elsewhere); page markers in the file supply page numbers.

Text checks. Documents with under 2,000 characters are flagged as likely scans. After the main run we found one article (Zaheer & Venkatraman, 1995) whose file was a first-page preview: 4,675 characters, one 107-word passage, effectively the abstract. Its claims scored as unusually coherent, which is how it surfaced. It was replaced with the full article from JSTOR (20 pages, 81,618 characters, cover page removed), re-extracted (8 passages), and every result that depended on it was rerun (A4). The extraction now also flags PDFs averaging under 1,000 characters per page.

Totals: 368 passages: legitimacy 184 (23 documents), TCE 148 (19), splice 36 (5). Re-extraction after the fix changed no other document's passages.

The repository stores passage IDs, pages, character offsets and SHA-256 hashes (`passage_index.csv`), never passage text. Anyone with the same PDFs regenerates identical text, and the hashes in `manifest_extraction.json` let them check.

A3. Readers, shams, claim extraction and editors

All prompts live in `prompts/` and were committed before any reading run. Every reader shares one template; only the method block and output schema differ, so each comparison changes one thing. The construct is "legitimacy" for the legitimacy corpus and "governance" for TCE and splice documents: TCE's central term, and the term the splice papers use in a corporate-governance sense.

Shared reader instructions (summary). You are one of several independent readers using different methods; you do not see the others' work. Read only the passages given, cite passage IDs for every point, and do not use outside knowledge of the document, its authors or the literature. Say where the passages are silent. Readers see a document ID, its year and its passages, never authors or titles.

The three methods (method blocks, verbatim).

- *Structural:* "For each claim this document makes about <construct>, ask what must be true for the claim to hold. Reconstruct the argument's architecture: the claims; the premises each claim depends on, whether stated or unstated; how the claims depend on one another; the scope conditions under which they are meant to hold; and any gaps, circularities, or unsupported steps. Do not judge whether the claims are empirically true…"
- *Operational:* "For each claim this document makes about <construct>, ask what one would observe if the claim were true. State the claims as testable propositions; identify the observable indicators each implies; describe the measurement strategy the document uses or implies; state what evidence would falsify each proposition; and note where <construct> cannot be observed or measured as the document states it…"
- *Genealogical:* "Ask how this document's understanding of <construct> came to be what it is. Identify the earlier formulations, authors, or traditions it draws on, revises, opposes, or replaces, as far as the passages show; what it retains from them; what it drops; what term or idea it puts in place of an earlier one; and what tensions inherited from earlier usage remain in it…"

Each method returns its own JSON object (premises and dependencies; propositions, indicators and falsifiers; lineage and substitutions) plus a reading of at most 150 words.

Shams. Two rewordings of the structural method with the same output schema: one plain-spoken ("Your job is to look at how the argument is put together…"), one formal and Latinate ("Adopt the posture of an analyst reconstructing argumentative architecture…"). Any divergence between shams can only come from wording.

Claim extraction (one prompt for every reader). Reduces a reading to 3–7 *characterization* claims (definition, components, mechanism, level of analysis, scope, and what the construct is distinguished from) and one *stance*. Each claim is one standalone declarative sentence of at most 25 words, in plain language, with no hedges and no method vocabulary (premise, proposition, falsify, indicator, lineage, substitution, reading, argument), carrying the passage IDs the reader cited. Stance labels: endorse, extend, modify, contest, reject, invoke; plus the account targeted.

Editors. Both editors receive, for every document, each real reader's stance, characterization claims and summary reading. They are told that disagreement in stance is not incoherence in characterization, and they are never told the hypotheses. Roles differ in one paragraph:

- *Synthesizing:* "Build the most coherent account of the literature that the readings support… record a point as contested only when it cannot be reconciled."
- *Adversarial:* "Find where the documents and readers genuinely disagree… Record a characterization as stable only if it survives a serious attempt to find documents or readers that depart from it."

The output schema is fixed: positions (sets of documents that characterize the construct in materially the same way, number left open), stable and contested characterizations with drivers (substantive, terminological, method, scope, unclear), reader disagreements, stance pattern, and an overall judgment: settled, coherent contestation, incoherent consensus or fragmentation.

A4. Runs

All model calls were made by `run_dcr.py` through the Claude Code command-line interface (`claude -p`) with model Claude Sonnet 5.5 (claude-sonnet-5-5, passed with --model). A test call with the same flag returned a usage report naming that model; the runner logged the requested model for each call but not Claude Code's per-call usage report, which Stage 2 will record. Each call replaced Claude Code's system prompt with ours (`--system-prompt`), was limited to one turn, and ran from an empty temporary folder, so the model had no files to read. Claude Code does not expose sampling parameters, so all calls used its defaults; the configured temperatures (1.0 for readers and editors, 0 for extraction) were not applied. Four calls ran in parallel; the runner resumes after interruption and never repeats a completed call.

Stage	Calls	Design
Reading	1,125	47 documents × 3 real readers × 5 repetitions, plus 42 documents × 2 shams × 5
Claim extraction	1,125	One per reading
Editing	42	2 editors × 3 conditions × 5 reader repetitions (30), plus 2

Stage	Calls	Design
		editors × 2 conditions × 3 repeats on repetition-1 input (12)

Outcomes. All 2,292 calls succeeded. 9 readings (0.8%) were not valid JSON and were passed to extraction as raw text. An audit (repair_runs.py --report) confirmed all nine were complete readings of 2,000–3,000 tokens with malformed JSON, not error messages, and that no call was recorded twice. 4 extractions (0.4%) did not parse; those readings have no claims and are excluded from the measures. Output totaled about 3.6 million tokens.

Pilot. A 15-call pilot (two legitimacy documents, all three stages) preceded the full run; every response parsed, and the readers produced method-distinct outputs.

Targeted rerun. After the corpus fix (A2), `invalidate_doc.py` moved every record that depended on the replaced document into `superseded.jsonl` (25 readings, 25 extractions, and the 26 editor runs for the TCE and spliced-TCE conditions), and the runner redid exactly those 76 calls. Original records remain in the repository and in Git history. The sensitivity run (A6) was interrupted and restarted at higher parallelism, and Claude Code then returned rate-limit errors inside otherwise successful responses, which the runner at first saved as readings. The runner now treats those as failures and retries them; repair_runs.py set aside 110 duplicate or empty records and the affected calls were redone, leaving one record per call and one parse failure (a complete reading). The same audit found no such records in the main run.

Logging. For each call, `runs_cc/requests/` records the model settings, SHA-256 hashes of both prompts and the Git commit, but not the prompt text, which contains copyrighted passages. `runs_cc/outputs/` records every response, its token usage and the parsed JSON.

A5. Measures

All measures are computed on characterization claims by `compute_measures.py`. Stance is analyzed separately and never counted as incoherence. NLI uses `microsoft/deberta-large-mnli` (He et al., 2021), scored in both directions; embeddings use `BAAI/bge-large-en-v1.5`, normalized.

Claim-pair score. For claims a and b, with entailment and contradiction probabilities averaged over both directions:

$$\mathrm{m}(\mathrm{a},\,\mathrm{b}) = \tfrac{1}{2}\,[\mathrm{P_e}(\mathrm{a}\to\mathrm{b}) + \mathrm{P_e}(\mathrm{b}\to\mathrm{a})] - \tfrac{1}{2}\,[\mathrm{P_c}(\mathrm{a}\to\mathrm{b}) + \mathrm{P_c}(\mathrm{b}\to\mathrm{a})]$$

Agreement between claim sets A and B (two readings, or two documents), ranging from −1 to 1:

$$\mathrm{S}(\mathrm{A},\,\mathrm{B}) = \tfrac{1}{2}\,[\mathrm{mean}_{\mathrm{a}\in\mathrm{A}}\,\mathrm{max}_{\mathrm{b}\in\mathrm{B}}\,\mathrm{m}(\mathrm{a},\,\mathrm{b}) + \mathrm{mean}_{\mathrm{b}\in\mathrm{B}}\,\mathrm{max}_{\mathrm{a}\in\mathrm{A}}\,\mathrm{m}(\mathrm{a},\,\mathrm{b})], \qquad \mathrm{D} = 1 - \mathrm{S}$$

Conflict ratio: contradiction pairs ÷ (contradiction + entailment pairs), counting a pair when its averaged probability exceeds 0.5. Neutral pairs, usually claims about different topics, are ignored.

Comparisons.

- *Noise floor:* same reader, same document, every pair of the 5 repetitions.
- *Real-reader divergence:* different real readers, same document, same repetition.

- *Sham control:* sham vs. sham, and sham vs. structural, same document and repetition.
- *Corpus coherence:* for each pair of documents within a corpus, S between their pooled real-reader claims, computed separately for each repetition. Each claim is compared with its 3 nearest claims in the other document by embedding similarity: cosine retrieves topically related claims; NLI judges them. Reported per condition (legitimacy, TCE, spliced TCE), with TCE–splice pairs broken out.
- *Editors:* judgment counts, positions, contested drivers, and agreement across the repeats on fixed input.
- *Co-citation baseline:* Spearman correlation between a document pair's mean S and its bibliographic coupling (Jaccard overlap of OpenAlex reference lists).

Within-document agreement uses all claim pairs. Confidence intervals resample documents with replacement (1,000 draws). Cosine versions of every agreement measure are reported as a secondary measure.

Gold-pair check (run before any reading). 16 constructed claim pairs, vetted by the author before scoring: 8 theoretically opposed but similarly worded, 8 equivalent but differently worded. In vetting, O4 (individual judgments vs. collective consensus) was dropped as compatible across levels of analysis, and O2 (property vs. perception) was sharpened into an explicit contradiction, leaving 15 pairs.

Measure	Separation (AUC)	Notes
NLI	1.00	14 of 15 labels correct; E7 ("bounded rationality" vs. "cannot foresee every contingency") labeled neutral
Cosine	0.18	Inverted: opposed pairs averaged 0.825, equivalent pairs 0.737

The gold set contained no pairs that are compatible but about different aspects of a construct. Its absence hid a bias suspected later (A8). A diagnostic of 10 such pairs (prompts/diagnostic_pairs.yaml, nli_diagnostic.py), committed before scoring and run after the main analysis, labeled 8 neutral, 2 contradictory and none entailment (mean P(contradiction) 0.20). Both misreadings contrast sibling categories in parallel wording: cognitive vs. pragmatic legitimacy (P = 0.67) and hierarchies vs. markets (0.58); perception vs. process came close (0.43). Because distinct but compatible claims almost never entail each other, even occasional misreadings dominate the conflict ratio, which explains its value of 0.65 within the noise floor.

A6. Results

Final measures after the targeted rerun (`results/report.md`). S is NLI agreement (−1 to 1); intervals resample documents.

Reader agreement within documents

Comparison	Pairs	S [95% CI]	Cosine	Conflict ratio
Noise floor, real readers (same reader, repeated)	1,394	0.325 [0.305, 0.343]	0.843	0.648
Noise floor, sham readers	840	0.323 [0.300, 0.344]	0.845	0.627
Sham vs. structural (same method)	420	0.319 [0.293, 0.342]	0.843	0.644
Sham vs. sham (same method)	210	0.309 [0.281, 0.337]	0.840	0.659
Sham vs. operational or genealogical	832	0.257 [0.236, 0.278]	0.813	0.737
Real vs. real (different methods)	697	0.256 [0.236, 0.277]	0.811	0.743

By pair of real readers: operational–structural 0.281 [0.258, 0.303]; genealogical–structural 0.251 [0.231, 0.271]; genealogical–operational 0.238 [0.215, 0.260]. By corpus (noise floor / real vs. real): legitimacy 0.338 / 0.270; TCE 0.301 / 0.230; splice 0.352 / 0.292.

Corpus coherence (agreement between documents, real readers pooled)

Condition	Document pairs	S (sd over 5 repetitions)	Cosine	Conflict ratio
Legitimacy	253	0.072 (0.003)	0.720	0.928
TCE	171	0.017 (0.004)	0.725	0.974
Splice papers among themselves	10	0.042 (0.009)	0.722	0.978
TCE spliced (all pairs)	276	−0.004 (0.004)	0.714	0.983
TCE–splice pairs only	95	−0.045 (0.005)	0.692	0.998

Stance (genealogical reader; modal label per document): legitimacy: modify 18, extend 5. TCE: extend 9, modify 7, contest 2, endorse 1. Splice: modify 3, extend 2.

Editors (5 reader repetitions per cell)

Editor	Condition	Judgments	Positions, mean (sd)	Contested drivers
Adversarial	Legitimacy	Coherent contestation 5	5.4 (0.5)	Substantive 18, scope 10
Adversarial	TCE	Coherent contestation 5	5.0 (0.0)	Substantive 20, scope 5, terminological 3
Adversarial	TCE spliced	Coherent contestation 5	6.2 (0.7)	Substantive 18, scope 8, terminological 2, method 1
Synthesizing	Legitimacy	Coherent	3.2 (1.8)	Substantive 8, scope 4, unclear

Editor	Condition	Judgments	Positions, mean (sd)	Contested drivers
		contestation 4, settled 1		2, terminological 1
Synthesizing	TCE	Coherent contestation 5	5.2 (0.4)	Substantive 15, scope 7, terminological 4
Synthesizing	TCE spliced	Coherent contestation 5	5.6 (0.5)	Substantive 17, scope 8, terminological 1

On fixed input (4 runs per editor and corpus), every editor repeated its judgment in all runs; positions varied by sd 0.43.

Co-citation baseline: Spearman ρ between document-pair agreement and bibliographic coupling = 0.43 (legitimacy, 253 pairs) and 0.26 (TCE with splice, 275 pairs).

Construct-sensitivity check

The TCE result could reflect the construct term rather than the literature: critics might redefine "governance" without disagreeing about TCE. We therefore reran the three real readers on the 19 TCE documents with the construct term "transaction cost economics" (`prompts/experiment_sensitivity.yaml`; 3 repetitions, 171 readings; no shams or editors; all prompts and models otherwise unchanged). The configuration and its predictions were committed before the run: if the split were an artifact of the term, coherence would rise toward legitimacy's 0.072; if not, it would stay near 0.017.

TCE construct term	Between-document S (sd over repetitions)	Within-document: noise floor / real vs. real
"governance" (main run, 5 repetitions)	0.017 (0.004)	0.301 / 0.230
"transaction cost economics" (3 repetitions)	0.025 (0.004)	0.332 / 0.229 [0.201, 0.259]

Coherence rose by 0.008, about two standard deviations, and stayed far below legitimacy (0.072): the split is not an artifact of the term. The method's core effect also replicated under the new term: real readers agreed below the noise floor, and the genealogical reader again agreed least with the others (0.211–0.215, vs. 0.262 for operational–structural). Two further observations, reported without interpretation: stance was similar (modify 9, extend 8, contest 1, endorse 1), and the correlation between document-pair agreement and bibliographic coupling fell to ρ = −0.06.

A7. Reproducibility, data and ethics

Order of commitments. Git timestamps show each rule fixed before the results it governs. The two later checks (the NLI diagnostic in A5 and the construct-sensitivity check in A6) were each committed, with

their predictions, before they ran. Changes after a freeze are logged in `prompts/CHANGELOG.md` with date and reason.

Date	Commit
2026-10-22	Inclusion rule and anchors frozen before the corpus build
2026-10-22	Corpus built and frozen (rule v1)
2026-10-24	Copyrighted text excluded from the repository
2026-10-25	PDF and extraction scripts; rule v2 logged as considered and rejected
2026-10-25	Corpus complete (47 of 47); passages extracted; book excerpts trimmed
2026-10-25	Gold pairs vetted (O4 dropped, O2 sharpened), before scoring
2026-10-25	Environment pinned; gold-pair check run (NLI AUC 1.00, cosine 0.18)
2026-10-25	Prompts, schemas and configuration frozen (TCE construct: governance)
2026-10-25	Runner committed
2026-10-26	Full run: 1,125 readings, 1,125 extractions, 42 editor runs
2026-10-26	Preview PDF replaced; targeted rerun; measures recomputed

What the repository holds.

Path	Contents
`select_corpus.py`, `fetch_pdfs.py`, `extract_passages.py`, `run_dcr.py`, `invalidate_doc.py`, `check_gold_pairs.py`, `compute_measures.py`	The full pipeline
`prompts/`	Reader, sham, extraction and editor prompts; `experiment.yaml`; gold pairs; changelog
`corpus/`	Selected and candidate lists with DOIs, reference lists, passage index with offsets and hashes, selection and extraction manifests
`runs_cc/`	Request logs (prompt hashes, settings, commits); extracted claims and editor judgments, including superseded records. Full reader outputs are private (see Release)
`results/`	Gold-pair check, report, per-pair CSVs, NLI cache
`requirements.txt`	Pinned Python environment (Python 3.11, PyTorch

Path	Contents
	2.14.1 + CUDA 12.6)

Not included: article PDFs, extracted text and passage text, which are copyrighted. A replicator obtains the PDFs by DOI, regenerates passages with `extract_passages.py`, and checks them against the stored hashes. Measures were computed on one laptop GPU (NVIDIA RTX 4070, 8 GB).

Release. The public repository holds the extracted claims, stances, editor judgments and all measures, which are enough to audit every reported number. Full reader outputs can quote short phrases from copyrighted passages, so they are kept in a private repository and available to the editors on request.

Ethics. The study uses published scholarship only; there are no human participants. Readers were blinded to authors and titles to limit reputation effects.

Anonymity for review. [If review is blind: share the repository through an anonymized mirror, and remove author names and email addresses from commit history and file metadata.]

A8. Known issues and Stage 2 fixes

Issue	Effect on results	Response
One model family, Claude Code defaults; temperature not controlled	Measured stability may partly reflect this model and its default sampling	Stage 2: rerun on the API with set temperatures, on a second Claude model and an open-weights model
NLI sometimes reads contrast as contradiction: repeated readings by one reader show a conflict ratio near 0.65	Deflates absolute agreement in every condition; comparisons hold because all conditions share the bias	Diagnosed (A5): 2 of 10 compatible pairs labeled contradictory, both contrasting sibling categories in parallel wording. Stage 2: calibrate or replace the NLI model, or merge such pairs before scoring
Claim extraction normalizes voice	Could make readings converge artificially	Bounded by the sham control and noise floor; Stage 2: compare against raw-reading measures
Stance is reliable only when a reader names the account engaged	Stance taken mainly from the genealogical reader	Stage 2: a short stance field for every reader
One PDF was a first-page preview	Its single-passage readings looked unusually coherent	Fixed and rerun (A2, A4); per-page text check added
Markets and Hierarchies passages are whole pages	Longer passages for one book	Accepted; noted
Editors label almost every corpus “coherent	Editor judgments are reliable but do not	Stage 2: finer editor categories, or editors anchored to the measures

Issue	Effect on results	Response
contestation”	discriminate corpora	
General-purpose NLI misses domain inferences (gold pair E7)	Some equivalent claims read as neutral	Stage 2: expert-labeled organizational-theory pairs
Readers may recognize famous papers despite blinding	Possible recall from memory	Stage 2: compare blinded readings of paraphrased passages
Small, curated corpus (47 documents)	Results are checkable, not general	Stage 2: 200+ documents per literature, more literatures